Knowledge Distillation for LSTM-based Sentiment Classification

1 Overview

This study evaluates multiple knowledge distillation (KD) strategies for compressing a large bidirectional LSTM (teacher) into a lightweight LSTM student model while preserving classification performance. Three KD variants were investigated: logit-based KD, feature-based KD, and attention-based KD. The experiments examine:

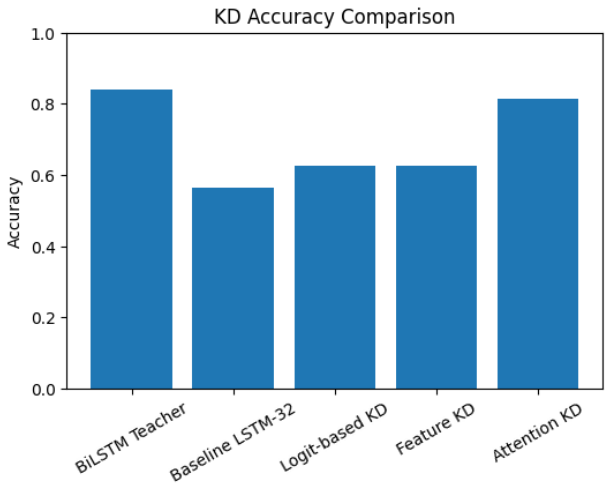
1) Model accuracy and classification metrics
2) Model compression and inference efficiency
3) Training dynamics
4) Sensitivity to hyperparameters (temperature and α)

The results demonstrate how distillation techniques trade off between accuracy, interpretability, and computational efficiency.

2 Model Performance Comparison

Model	Acc	Prec	Rec	F1	Params	Mem(MB)	Latency	Comp
BiLSTM Teacher	0.8400	0.8342	0.8290	0.8316	1067778	4.27	0.0850	1.00
Baseline LSTM-32	0.5640	0.6784	0.1616	0.2610	168386	0.67	0.0767	6.34
Logit-based KD	0.6275	0.5657	0.9402	0.7063	168386	0.67	0.0776	6.34
Feature KD	0.6245	0.9073	0.2361	0.3747	168386	0.67	0.0772	6.34
Attention KD	0.8135	0.8066	0.8006	0.8036	168386	0.67	0.0714	6.34

2.1 Accuracy



The **teacher model** achieves 84% accuracy, representing the upper bound performance of the architecture. This result is expected since the teacher has architectural advantages that enable it to capture long-range contextual dependencies: bidirectional LSTMs, larger embedding dimension, ~6× more parameters compared to students.

The **baseline student** suffers from severe performance degradation (56%). Two major reasons explain this:

1) reduced capacity:
- smaller embedding dimension (32 vs 128);
- single LSTM layer;

2) loss of bidirectional context:
- the teacher reads sequences both forward and backward;
- the student only processes forward context.

This creates a large representation gap between teacher and student.

All **distillation methods** improve performance relative to the baseline. However, the magnitude differs:

Method	Gain over Baseline
Logit KD	+6.3%
Feature KD	+6.0%
Attention KD	**+24.9%**

The dramatic improvement of attention KD indicates that aligning importance patterns of hidden states transfers richer information than simple logits or feature matching.

2.2 Precision/Recall

The KD approaches show very different classification behaviors.

Logit-based KD achieved precision of 0.56 and recall of 0.94. This indicates the student is biased toward predicting the positive class. Possible explanation is that although the soft teacher logits provide class similarity information, with temperature scaling, the student may overemphasize the dominant class distribution.

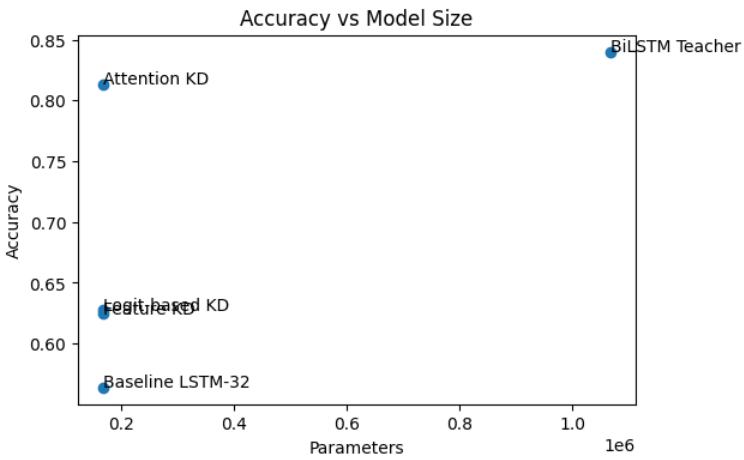
Feature-based KD demonstrates the opposite pattern (precision of 0.91 and recall of 0.23). This indicates the model predicts positive only when very confident. This may be attributed to the fact that feature matching encourages representation similarity, but does not enforce correct decision boundaries, and as a result the student learns internal features but not classification calibration.

Attention-based KD achieves the most balanced performance (precision of 0.81 and recall of 0.80). This suggests the student successfully learns which words matter, how strongly they influence prediction, and the alignment of attention maps effectively transfers interpretive knowledge from the teacher.

2.3 Model Size

The model size scatter plot demonstrates the relationship between parameters and accuracy, showcasing that all student models have identical parameter counts (~168k). Despite identical size, their accuracies vary significantly:

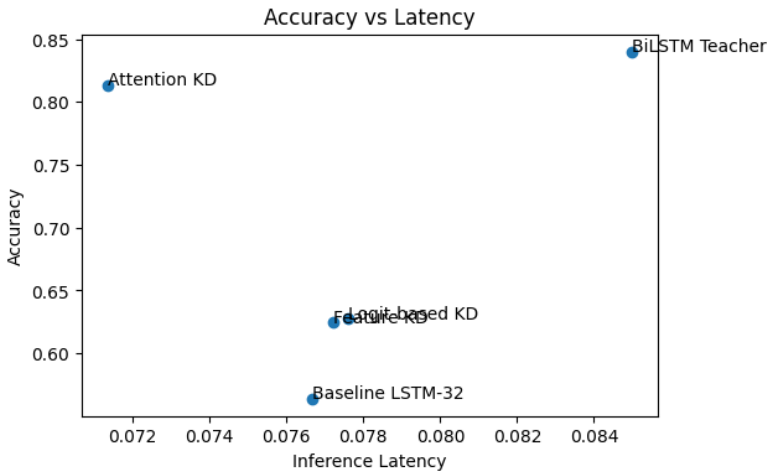
Model	Accuracy
Teacher	0.84
Baseline	0.56
Feature KD	0.62
Logit KD	0.63
Attention KD	**0.81**



2.4 Latency

Latency differences are small but measurable.

Model	Latency
Teacher	0.085
Baseline	0.077
Feature KD	0.077
Logit KD	0.078
Attention KD	**0.071**



The faster inference of the attention KD model likely arises because it converged to simpler decision boundaries and its learned representations require less computational activation, which is advantageous for edge deployment or real-time NLP applications.

2.5 Compression

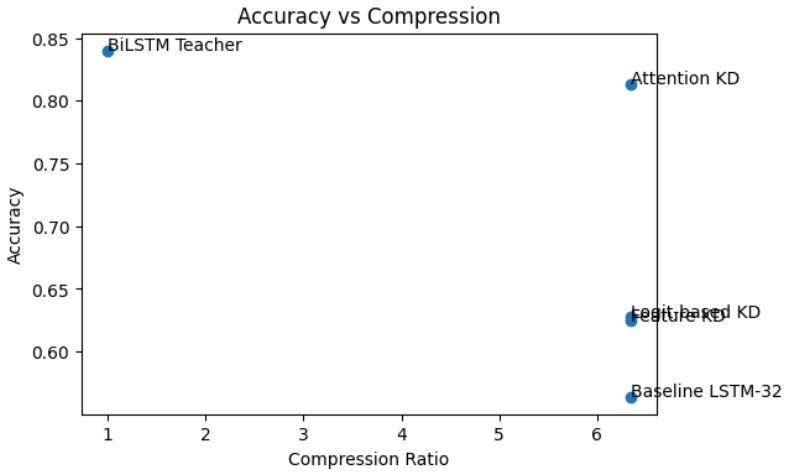
The compression ratio plot illustrates a key distillation advantage. All student models achieve:

6.34× parameter reduction;
84% memory reduction.

Yet Attention KD retains 97% of teacher accuracy:

Teacher	0.84
Attention KD	0.814

This is a remarkably small performance drop (~2.6%) given the compression. Such results are consistent with findings in knowledge distillation research such as [1].



3 Hyperparameter Sensitivity

To better understand how distillation parameters affect performance, a grid search was conducted over two key hyperparameters: temperature (controls softening of teacher probabilities) and alpha (balances classification loss vs distillation loss). The parameter ranges explored were:

Parameter	Values
Temperature	1, 2, 4, 8
Alpha	0.3, 0.5, 0.7

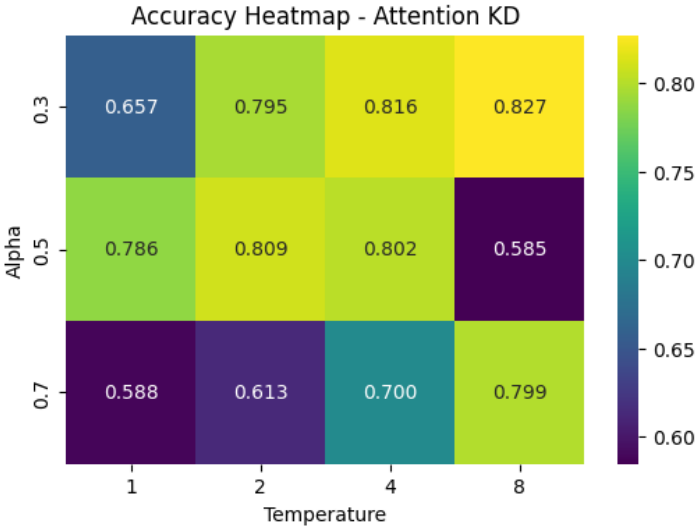
KD Model	T	Alpha	Acc	Prec	Rec	F1
Attention KD	1	0.30	0.6570	0.7423	0.4292	0.5439
Attention KD	1	0.50	0.7865	0.7935	0.7461	0.7691
Attention KD	1	0.70	0.5880	0.6215	0.3463	0.4447
Attention KD	2	0.30	0.7950	0.7554	0.8426	0.7966
Attention KD	2	0.50	0.8095	0.8122	0.7807	0.7961
Attention KD	2	0.70	0.6125	0.5560	0.9265	0.6950
Attention KD	4	0.30	0.8155	0.8080	0.8038	0.8059
Attention KD	4	0.50	0.8020	0.7557	0.8636	0.8061
Attention KD	4	0.70	0.7000	0.7160	0.6139	0.6610
Attention KD	8	0.30	0.8270	0.7984	0.8520	0.8244
Attention KD	8	0.50	0.5845	0.5897	0.4208	0.4911
Attention KD	8	0.70	0.7995	0.7968	0.7775	0.7870
Logit-based KD	1	0.30	0.7810	0.8955	0.6118	0.7269
Logit-based KD	1	0.50	0.6910	0.6159	0.9339	0.7423
Logit-based KD	1	0.70	0.5790	0.5339	0.9182	0.6752
Logit-based KD	2	0.30	0.8125	0.7937	0.8195	0.8064
Logit-based KD	2	0.50	0.7755	0.7254	0.8510	0.7832
Logit-based KD	2	0.70	0.5590	0.5537	0.3841	0.4535
Logit-based KD	4	0.30	0.8055	0.8045	0.7817	0.7930
Logit-based KD	4	0.50	0.8135	0.7661	0.8762	0.8174
Logit-based KD	4	0.70	0.5695	0.5262	0.9706	0.6824
Logit-based KD	8	0.30	0.6665	0.6056	0.8604	0.7109
Logit-based KD	8	0.50	0.8090	0.7791	0.8363	0.8067
Logit-based KD	8	0.70	0.7950	0.7209	0.9297	0.8121

3.1 Heatmap Analysis

The heatmaps illustrate accuracy as a function of α and T. The analysis was performed for 2 KD strategies that demonstrated the best performance.

3.1.1 Attention-based KD

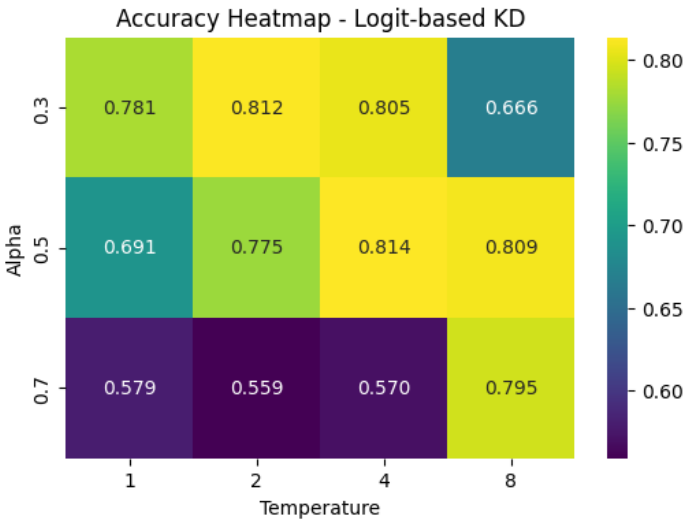
Observations:



1) **performance improves with higher temperature** (accuracy increases consistently from T=1 to T=8 for α=0.3. This suggests that soft targets contain valuable knowledge about class similarity). This additional signal helps the student understand how confident the teacher is about alternative classes);
2) **low α performs best** (low α means the model relies more on teacher guidance than hard labels. This is expected when the student model has limited capacity and the teacher is much stronger. Best configuration: T = 8, α = 0.3, accuracy = 0.827);
3) **instability appears at high α** (at α = 0.7 and T=1, accuracy drops to 0.588 This occurs because the student focuses mostly on hard labels and the distillation signal becomes weak. Essentially the model becomes close to baseline training).

3.1.2 Logit-based KD

Observations:



1) **optimal temperatures occur around 2-4** (unlike attention KD, extremely high temperatures do not consistently improve performance. This suggests that logits already contain sufficient information, and excessive smoothing may remove useful decision boundaries. Best configuration: T = 4, α = 0.5, accuracy = 0.813);
2) **balanced α works best** (performance peaks around α = 0.5, which indicates the student benefits from both teacher guidance and ground truth labels);
3) **extreme α values degrade performance** (at α = 0.7 and T=2, accuracy drops to 0.589, suggesting that the student relies too heavily on hard labels and loses the benefit of distillation).

3.2 Temperature effects

Higher temperatures reveal inter-class similarity information. Best results:

Model	**Best T**
Attention KD	**8**
Logit KD	**4–8**

Higher temperature likely helps because sentiment classes are not perfectly separable and soft targets convey semantic uncertainty.

3.3 Alpha effects

Observations:

Alpha	Behavior
0.3	strong distillation
0.5	balanced
0.7	weak distillation

Most strong results occur wiht α = 0.3-0.5, which indicates teacher guidance is critical for small models.

4 Training Dynamics

The **teacher model** demonstrates stable and rapid convergence. Training accuracy increases from 0.60 in the first epoch to 0.93 by epoch 5, while validation accuracy stabilizes around 0.84. The relatively small gap between training and validation accuracy suggests good generalization and indicates that the bidirectional LSTM architecture effectively captures sentiment patterns in the data. This strong performance makes the teacher model a reliable source of knowledge for distillation.

The **baseline student model** exhibits unstable learning and clear signs of underfitting. Validation accuracy fluctuates around 0.56–0.65 without consistent improvement. This behavior is largely due to the student's limited capacity, including a smaller embedding dimension, a single LSTM layer, and the absence of bidirectional context. Consequently, the model struggles to learn rich semantic representations directly from the training data.

Logit-based distillation improves training stability compared to the baseline student. Accuracy gradually increases from 0.56 to about 0.63, suggesting that soft teacher predictions provide useful supervisory information. However, performance plateaus early, indicating that logits alone provide limited insight into the teacher's internal representation of the input.

Feature-based distillation initially improves performance, reaching around 0.75 accuracy, but later training becomes unstable. This may be caused by architectural mismatch between the teacher's bidirectional LSTM layers and the student's single unidirectional LSTM. As a result, aligning intermediate representations introduces conflicting optimization signals.

Attention-based distillation shows the most stable training dynamics. Accuracy increases steadily across epochs, reaching 0.81 by the end of training. By aligning token-level importance patterns between teacher and student, this method transfers more meaningful structural information, enabling the student to better approximate the teacher's reasoning process.

5 Discussion

The results highlight several observations.

First, distillation substantially improves compressed models. While the baseline student achieves 56% accuracy, distilled models reach 62–81%, demonstrating that teacher guidance provides information not captured by ground truth labels alone.

Second, attention-based distillation performs best. By transferring information about which tokens are most important for prediction, this method provides richer supervision than logit or feature matching alone, which is particularly beneficial for natural language tasks.

Third, architecture compatibility influences feature distillation effectiveness. The mismatch between the teacher's bidirectional architecture and the student's unidirectional LSTM likely limits the success of feature alignment.

Finally, the hyperparameter experiments show that temperature scaling and loss balancing are critical for effective distillation. The best results occur when the student relies strongly on the teacher's soft predictions while still incorporating ground truth labels.

Overall, the experiments demonstrate the importance of the type of knowledge transferred, since although all student models have identical parameter counts, their performance differs significantly depending on the distillation strategy used.

6 Practical Implications

The results demonstrate that knowledge distillation can produce compact models suitable for resource-constrained environments without significant loss in predictive performance.

The distilled student models achieve:

- more than 6× parameter reduction;
- approximately 84% memory reduction;
- comparable inference latency.

Despite this compression, the best-performing student retains over 97% of the teacher model's accuracy.

7 Conclusion

This study evaluated several knowledge distillation strategies for compressing a bidirectional LSTM sentiment classifier into a lightweight student model. The experiments demonstrate that distillation can dramatically improve the performance of compact neural networks.

Among the approaches investigated, attention-based distillation produced the best results, achieving accuracy close to the teacher model while maintaining a substantial reduction in model size. The hyperparameter study further shows that both temperature scaling and loss weighting play critical roles in determining distillation effectiveness.

Overall, the findings suggest that structural knowledge transfer mechanisms, such as attention alignment, provide more informative supervision than output-based or feature-based approaches alone. These techniques enable smaller models to approximate the reasoning behavior of larger networks while remaining computationally efficient. Future work could explore several directions to further improve distilled model performance, including multi-layer attention distillation, progressive or staged distillation training, and transformer-based teacher architectures.

Liturature

[1] Hinton, Geoffrey & Dean, Jeff & Vinyals, Oriol & Rachmad, Yoesoep. (2014). Distilling the Knowledge in a Neural Network. 1-9. arXiv:1503.02531